ktu

1922

T171B101 Lustų ir jų sistemų prototipavimo, gamybos ir testavimo technologijos

80% of Git

Initial Setup

```
# Option 1:
# do once on your machine
git config --global user.name "MlabUser"
git config --global user.email "mlab.user@ktu.edu"


# the "unsafe" way to store credentials, use github auth tokens (classic)
# use before cloning or pulling/pushing to remote repository
git config --global credential.helper store


# Option 2:
# use github cli: https://cli.github.com/
gh auth login
```

Project Initialization

```
# clone already existing project
git clone <repo>

# initialize your own
git init
```

Daily Driver

```
# check the state of a project files
git status
git status -s


# stage a specific file
git add sw/hal/spi.c


# stage all files in the current working directory
git add .


# commit staged file changes
git commit -m "feat(spi): full working spi_read"
```

Staging

Git

Action

```
git status -s
```

Project

```
└── readme.md
```

Git Action Project

```
.
├── Makefile
├── readme.md
└── src
    └── main.c
```

Git

```
?? Makefile
?? src/
```

Action

```
git status -s
```

Project

```
.
├── Makefile
├── readme.md
└── src
    └── main.c
```

Git

```
?? Makefile
?? src/
```

Action

```
git add Makefile
```

Project

```
.
├── Makefile
├── readme.md
└── src
    └── main.c
```

Git

```
A   Makefile
?? src/
```

Action

```
git status -s
```

Project

```
.
├── Makefile
├── readme.md
└── src
    └── main.c
```

Git

```
A  Makefile
?? src/
```

Action

```
git add src/
```

Project

```
.
├── Makefile
├── readme.md
└── src
    └── main.c
```

Git

```
A  Makefile
A  src/main.c
```

Action

```
git status -s
```

Project

```
.
├── Makefile
├── readme.md
└── src
    └── main.c
```

Git

Action

Project

```
.
├── Makefile
└── src
    └── main.c
```

Git

```
A   Makefile
 D  readme.md
A   src/main.c
```

Action

```
git status -s
```

Project

```
.
├── Makefile
└── src
    └── main.c
```

Git

```
A   Makefile
 D  readme.md
A   src/main.c
```

Action

```
git add .
```

Project

```
.
├── Makefile
└── src
    └── main.c
```

Git

```
A   Makefile
D   readme.md
A   src/main.c
```

Action

```
git status -s
```

Project

```
.
├── Makefile
└── src
    └── main.c
```

Git	Action	Project
A Makefile D readme.md A src/main.c	`git restore --staged Makefile`	. ├── Makefile └── src └── main.c

Git

```
?? Makefile
D  readme.md
A  src/main.c
```

Action

```
git status -s
```

Project

```
.
├── Makefile
└── src
    └── main.c
```

Git

```
?? Makefile
D  readme.md
A  src/main.c
```

Action

```
git restore --staged .
```

Project

```
.
├── Makefile
└── src
    └── main.c
```

Git

```
 D readme.md
?? Makefile
?? src/
```

Action

```
git status -s
```

Project

```
.
├── Makefile
└── src
    └── main.c
```

Git

```
 D readme.md
?? Makefile
?? src/
```

Action

```
git add .
```

Project

```
.
├── Makefile
└── src
    └── main.c
```

Git

```
D readme.md
A Makefile
A src/
```

Action

```
git status -s
```

Project

```
.
├── Makefile
└── src
    └── main.c
```

Committing

Git Log

Action

Git Log

Action

```
git commit -m "init: project"
```

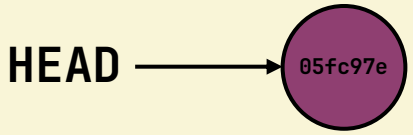
Git Log

Action

```
git commit -m "init: project"
```

Git Log

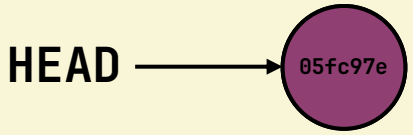
Action



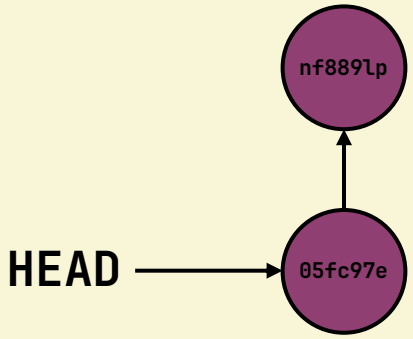
Git Log

Action

```
git commit -m "chore: Makefile to load program"
```



Git Log

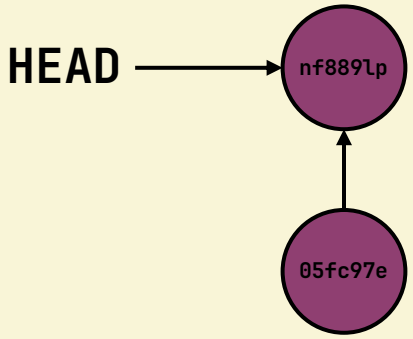


Action

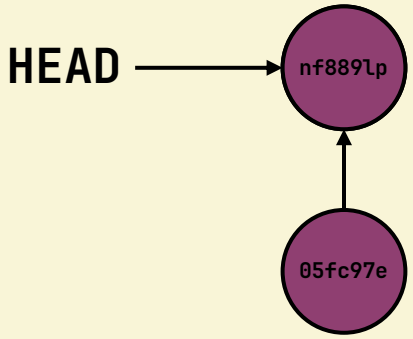
```
git commit -m "chore: Makefile to load program"
```

Git Log

Action



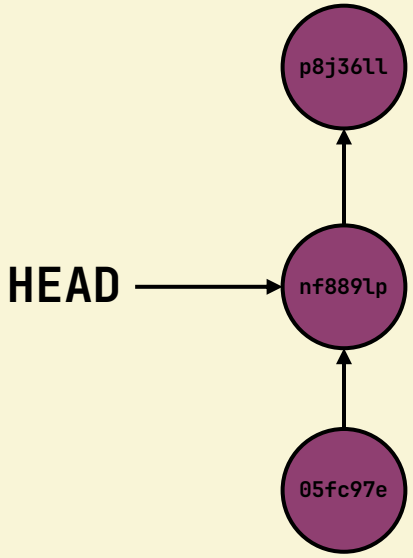
Git Log



Action

```
git commit -m "feat: heartbeat program"
```

Git Log

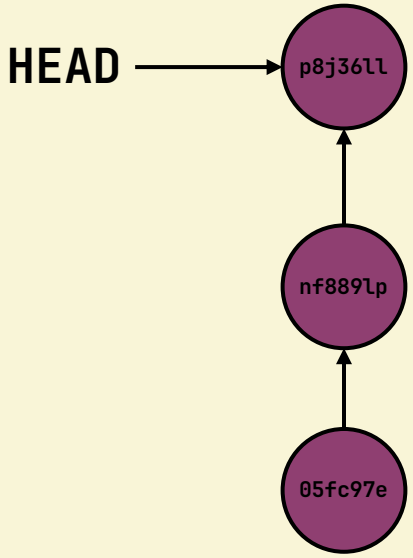


Action

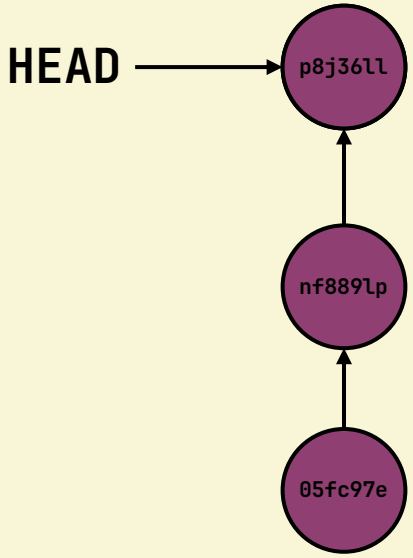
```
git commit -m "feat: heartbeat program"
```

Git Log

Action



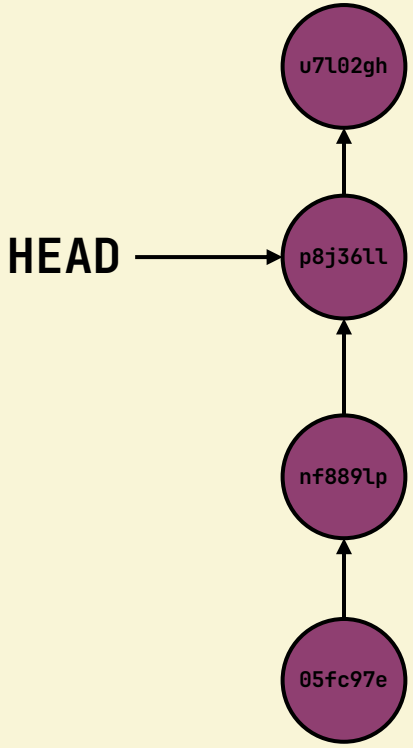
Git Log



Action

```
git commit -m "refac: cleared code structure"
```

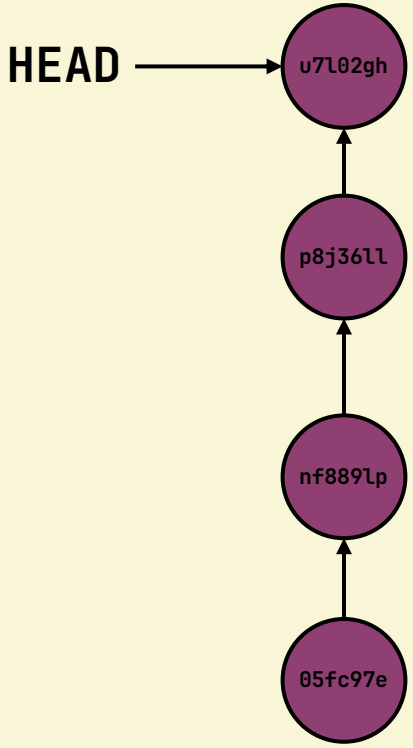
Git Log



Action

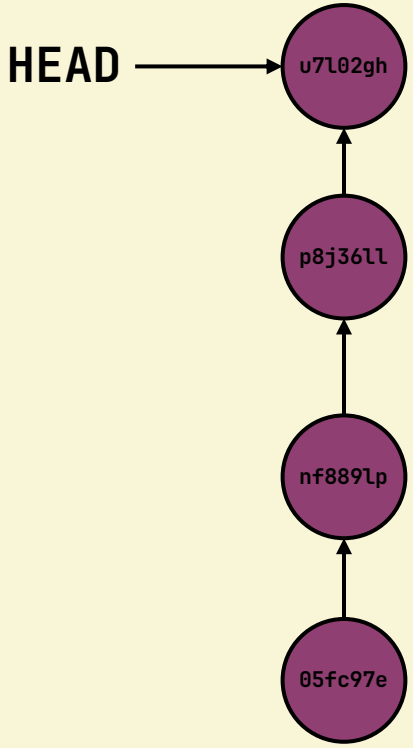
```
git commit -m "refac: cleared code structure"
```

Git Log



Action

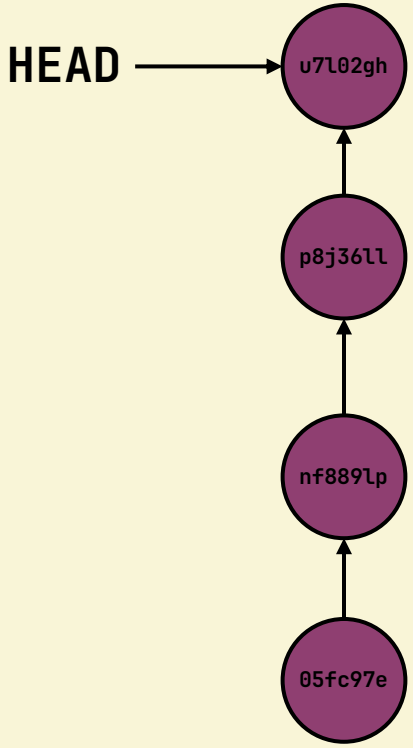
Git Log



Action

```
git log --oneline
```

Git Log

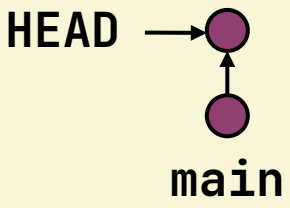



```
u7l02gh (HEAD -> main) refac: cleared code structure
p8j36ll feat: heartbeat program
nf889lp chore: Makefile to load program
05fc97e init: project
```

Branching

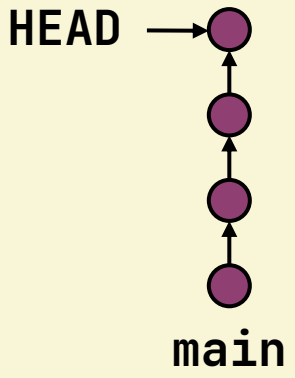
Git Log

Action

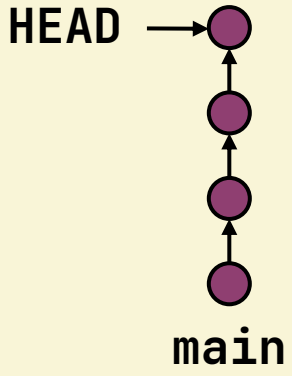


Git Log

Action



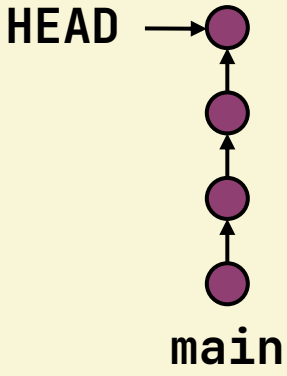
Git Log



Action

```
git branch dev
```

Git Log



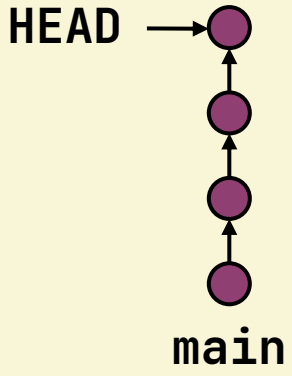
Action

```
git branch -a
```

Git

```
  dev
* main
```

Git Log



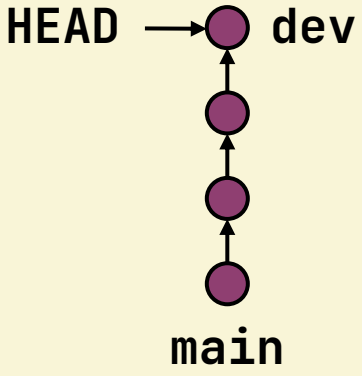
Action

```
git checkout dev
```

Git

```
Switched to branch 'dev'
```

Git Log



Action

```
git branch -a
```

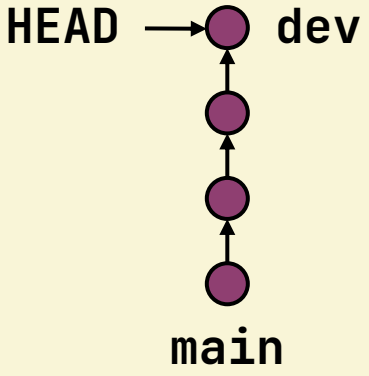
Git

```
* dev
  main
```

Git Log

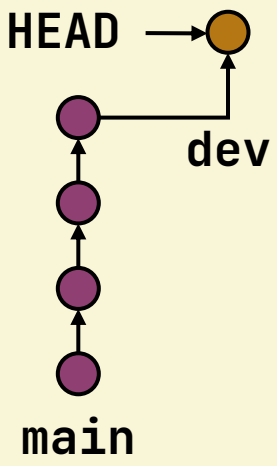
Action

```
git log --oneline --graph
```




```
* 059c012 (HEAD -> dev, main) refac: cleared code structure
* 18dcb30 feat: heartbeat program
* 7db895b chore: Makefile to load program
* c03be49 init: project
```

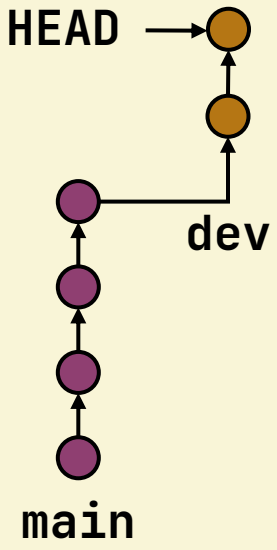
Git Log



Action

```
git commit -m "wip: spi driver"
```

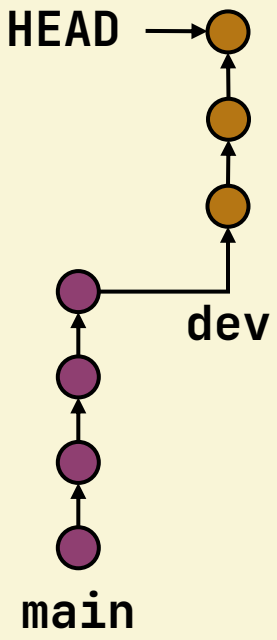
Git Log



Action

```
git commit -m "feat: small spi driver"
```

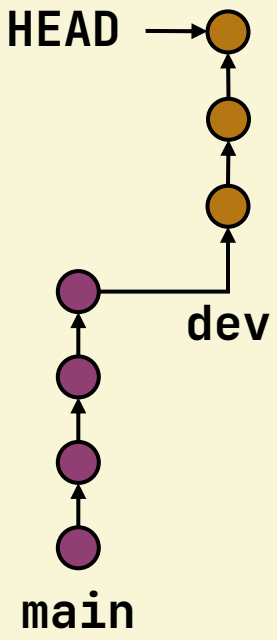
Git Log



Action

```
git commit -m "fix: spi write irq stall"
```

Git Log



Action

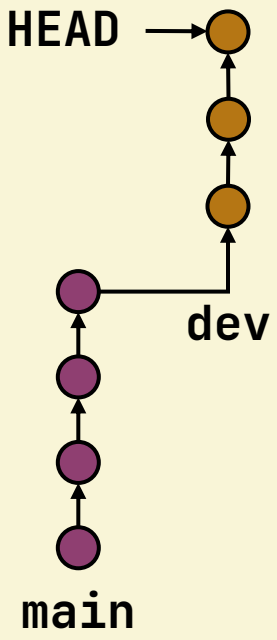
```
git log --oneline --graph
```

```
* e30a8ea (HEAD -> dev) fix: spi write irq stall
* e3e1ead feat: small spi driver
* cc09547 wip: spi driver
* 059c012 (main) refac: cleared code structure
* 18dcb30 feat: heartbeat program
* 7db895b chore: Makefile to load program
* c03be49 init: project
```

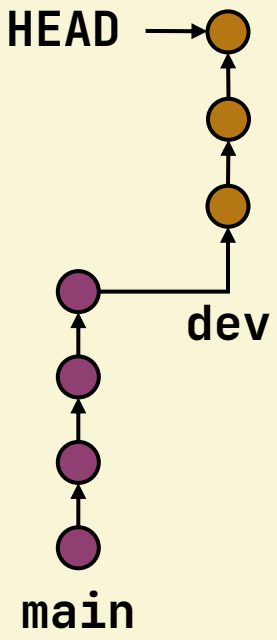
Rebasing

Git Log

Action



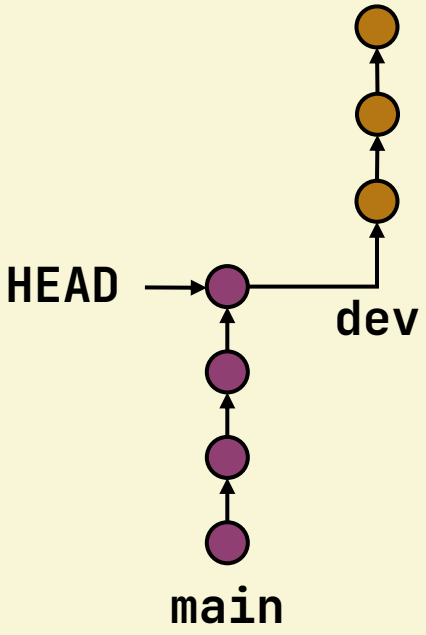
Git Log



Action

```
git checkout main
```

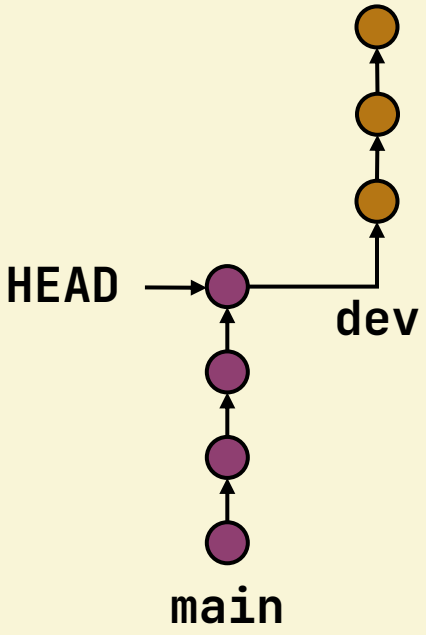
Git Log



Action

```
git checkout main
```

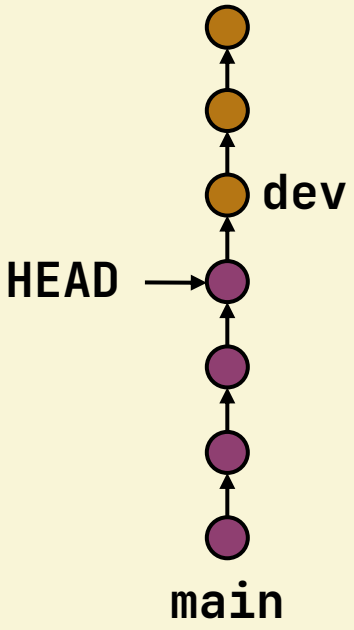
Git Log



Action

```
git rebase dev
```

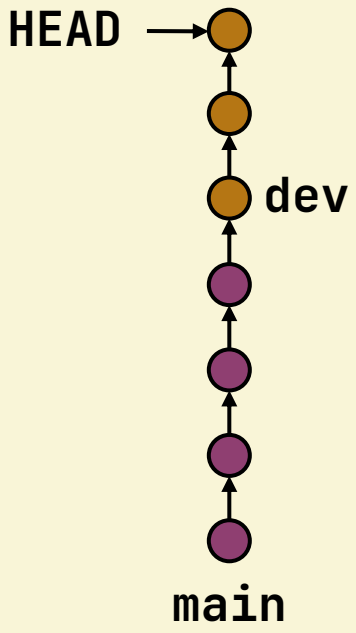
Git Log



Action

```
git rebase dev
```

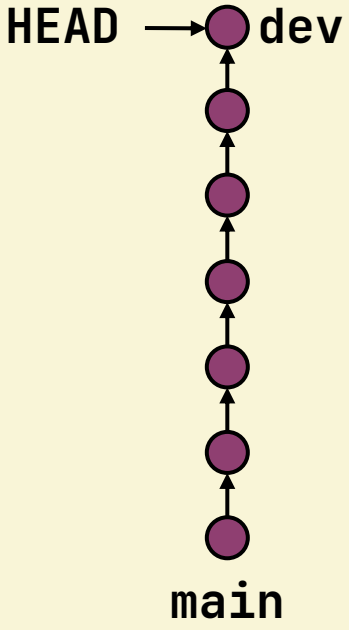
Git Log



Action

```
git rebase dev
```

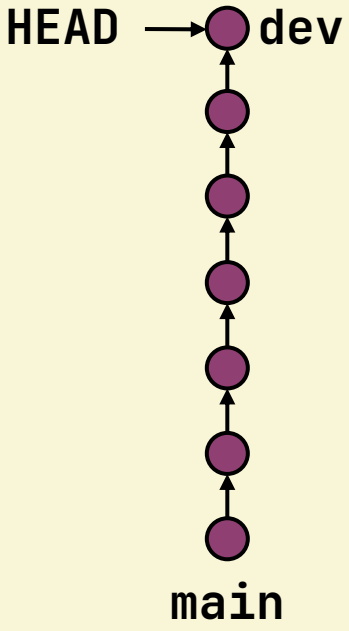
Git Log



Action

```
git rebase dev
```

Git Log



Action

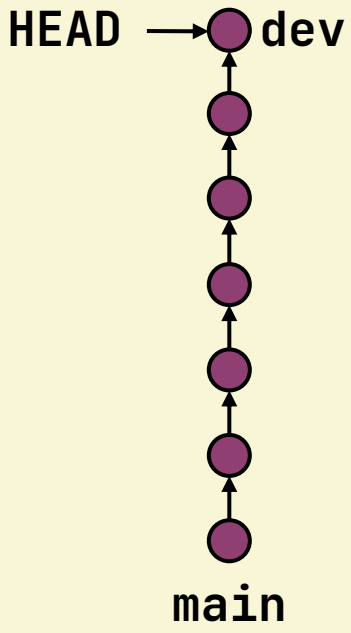
```
git rebase dev
```

```
Successfully rebased and updated refs/heads/main.
```

Git Log

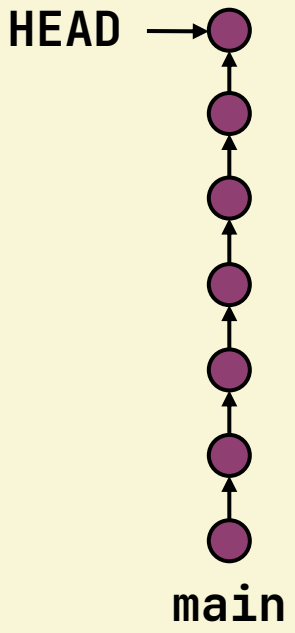
Action

```
git log --oneline --graph
```




```
* e30a8ea (HEAD -> main, dev) fix: spi write irq stall
* e3e1ead feat: small spi driver
* cc09547 wip: spi driver
* 059c012 refac: cleared code structure
* 18dcb30 feat: heartbeat program
* 7db895b chore: Makefile to load program
* c03be49 init: project
```

Git Log



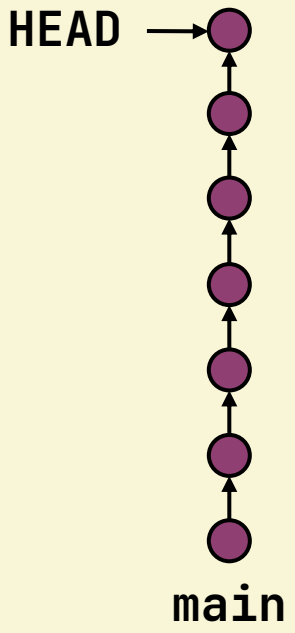
Action

```
git branch -d dev
```

```
Deleted branch dev (was e30a8ea).
```

Merging

Git Log

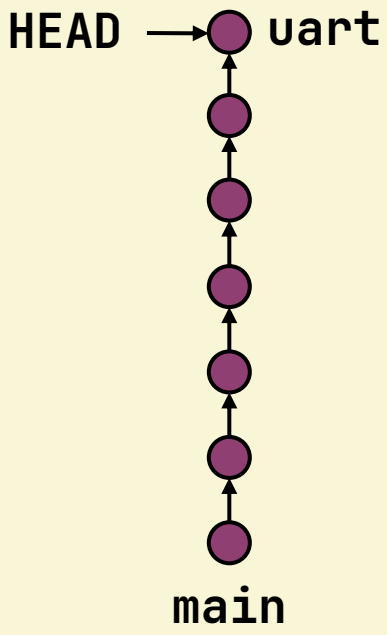


Action

```
git checkout -b uart
```

```
Switched to a new branch 'uart'
```

Git Log

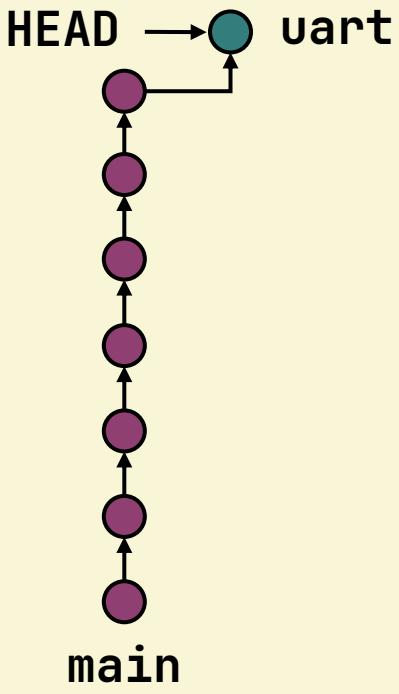


Action

```
git checkout -b uart
```

```
Switched to a new branch 'uart'
```

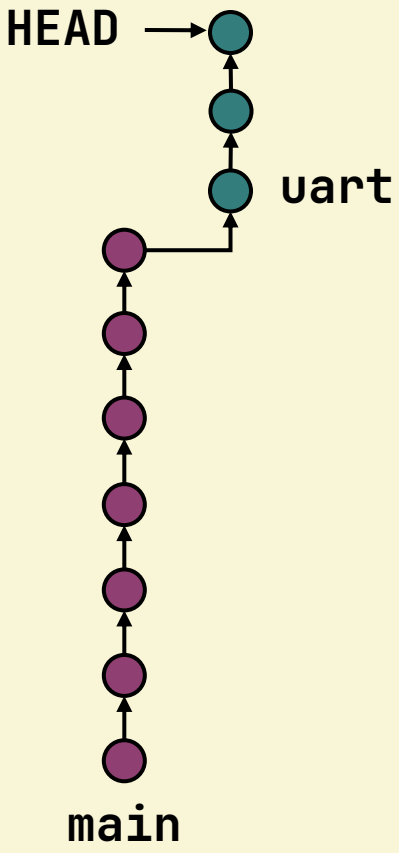
Git Log



Action

```
git commit -am "wip: uart driver"
```

Git Log

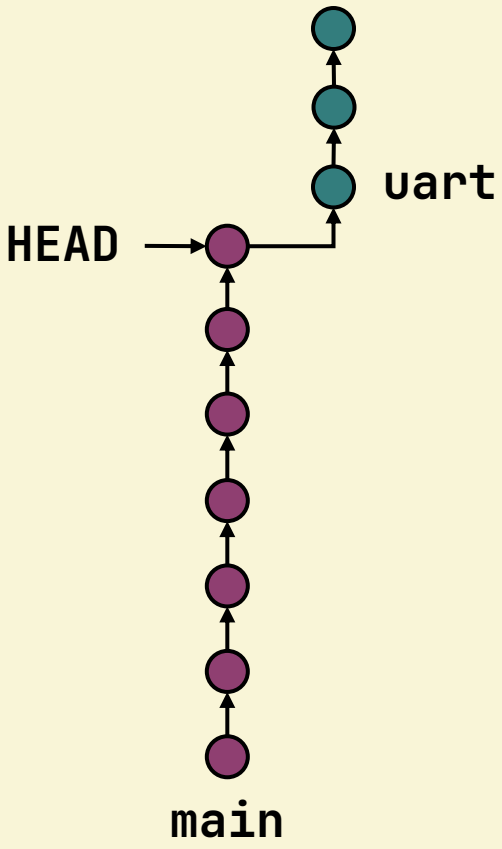


Action

```
git commit -am "wip: uart driver“
...
git commit -am "feat: fully working uart"
```

```
* df7e276 (HEAD -> uart) test: uart driver
* c9d11ab feat: fully working uart
* d4b7727 doc: add to uart driver source code doxygen
* 64d1c87 chore: update Makefile
* c4c8c2c feat: simple uart driver
* 25ef1cc wip: uart driver
* e30a8ea (main) fix: spi write irq stall
```

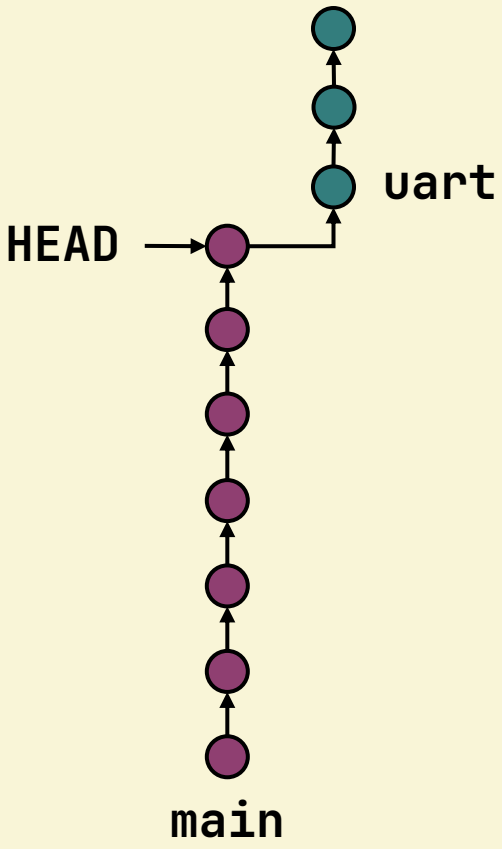
Git Log



Action

```
git checkout main
```

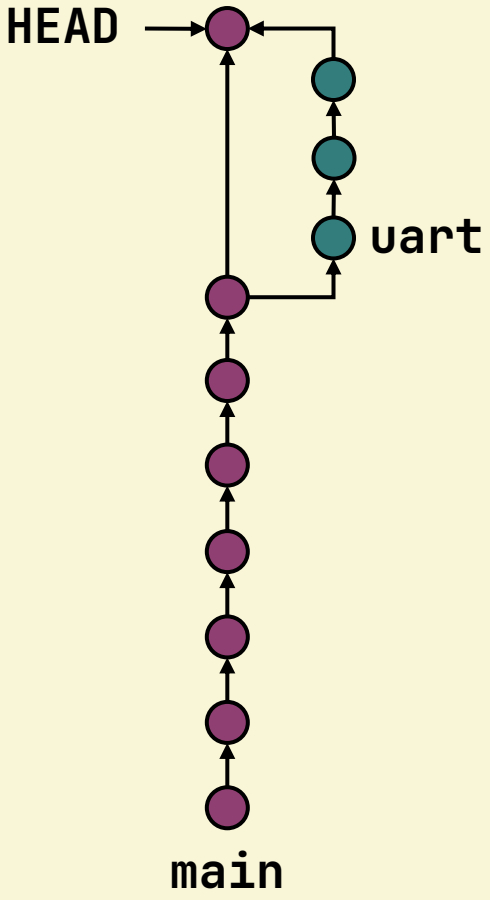
Git Log



Action

```
git merge uart
```

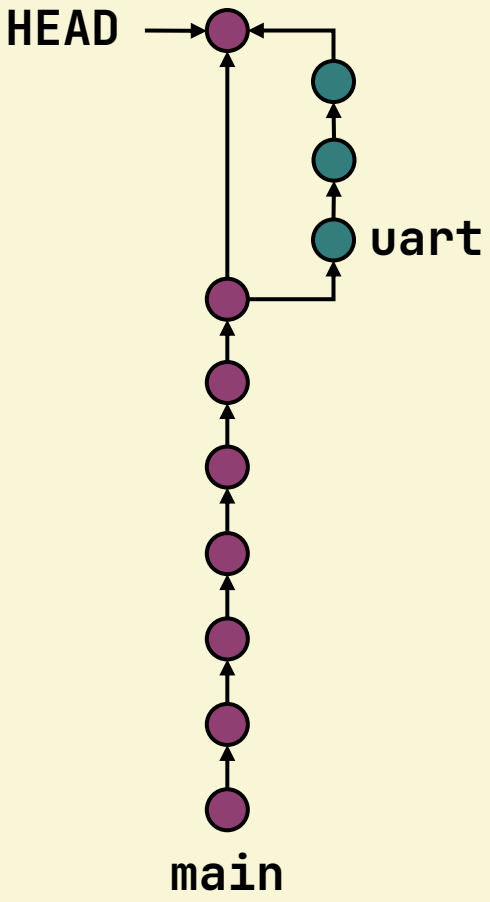
Git Log



Action

```
git merge uart
```

Git Log



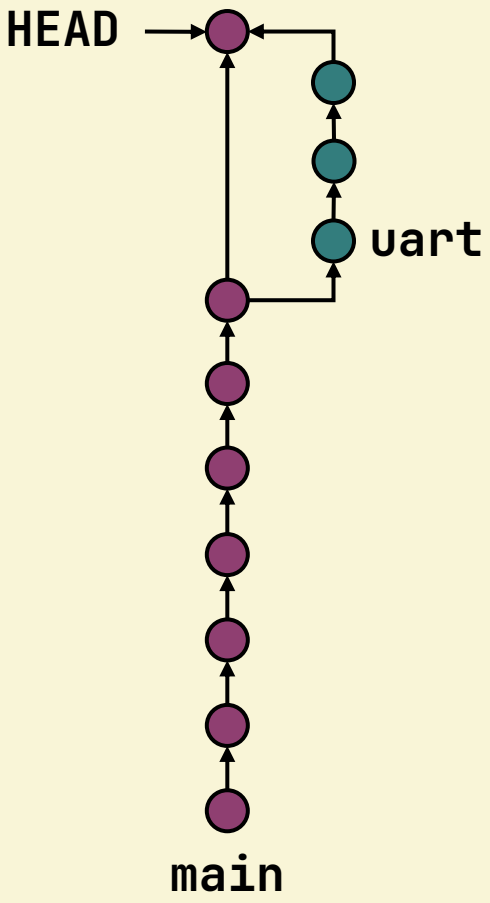
Action

```
git log --oneline --graph --all
```

```
*   77bb1ee (HEAD -> main) merge: branch 'uart' to 'main'
|\
| * df7e276 (uart) test: uart driver
| * c9d11ab feat: fully working uart
| * d4b7727 doc: add to uart driver source code doxygen
| * 64d1c87 chore: update Makefile
| * c4c8c2c feat: simple uart driver
| * 25ef1cc wip: uart driver
|/
* e30a8ea fix: spi write irq stall
```

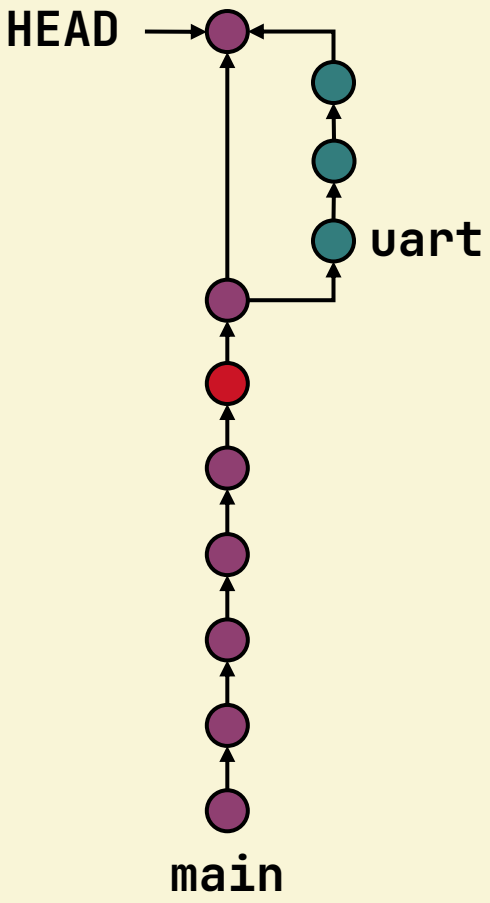
Checkouting

Git Log



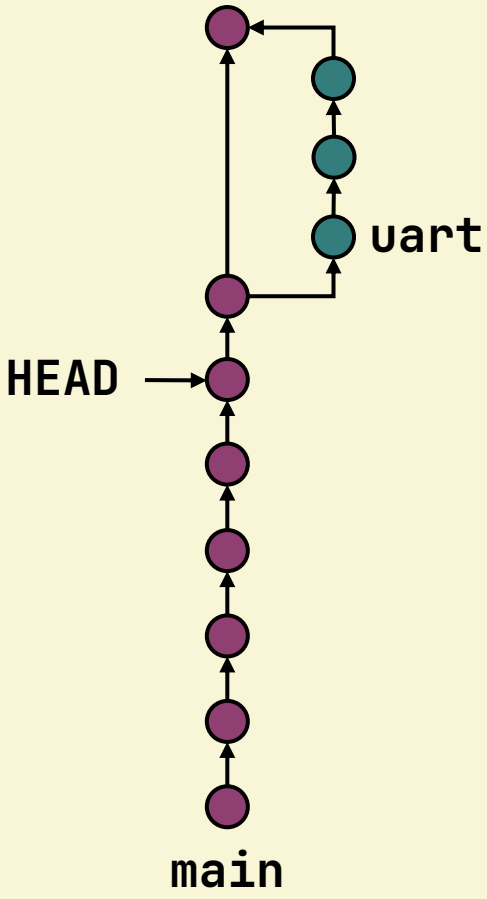

```
*   77bb1ee (HEAD -> main) merge: branch 'uart' to 'main'
|\
| * df7e276 (uart) test: uart driver
| * c9d11ab feat: fully working uart
| * d4b7727 doc: add to uart driver source code doxygen
| * 64d1c87 chore: update Makefile
| * c4c8c2c feat: simple uart driver
| * 25ef1cc wip: uart driver
|/
* e30a8ea fix: spi write irq stall
* e3e1ead feat: small spi driver
* cc09547 wip: spi driver
* 059c012 refac: cleared code structure
* 18dcb30 feat: heartbeat program
* 7db895b chore: Makefile to load program
* c03be49 init: project
```

Git Log



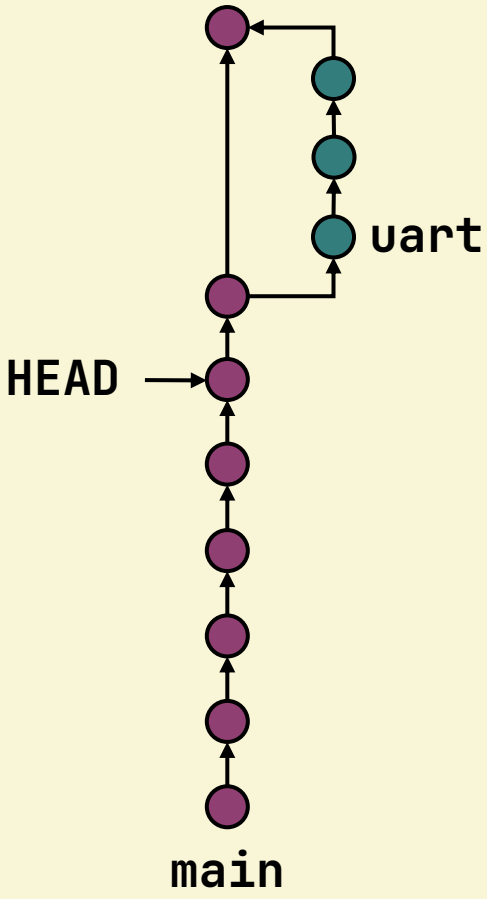

```
*   77bb1ee (HEAD -> main) merge: branch 'uart' to 'main'
|\
| * df7e276 (uart) test: uart driver
| * c9d11ab feat: fully working uart
| * d4b7727 doc: add to uart driver source code doxygen
| * 64d1c87 chore: update Makefile
| * c4c8c2c feat: simple uart driver
| * 25ef1cc wip: uart driver
|/
* e30a8ea fix: spi write irq stall
* e3e1ead feat: small spi driver
* cc09547 wip: spi driver
* 059c012 refac: cleared code structure
* 18dcb30 feat: heartbeat program
* 7db895b chore: Makefile to load program
* c03be49 init: project
```

Git Log



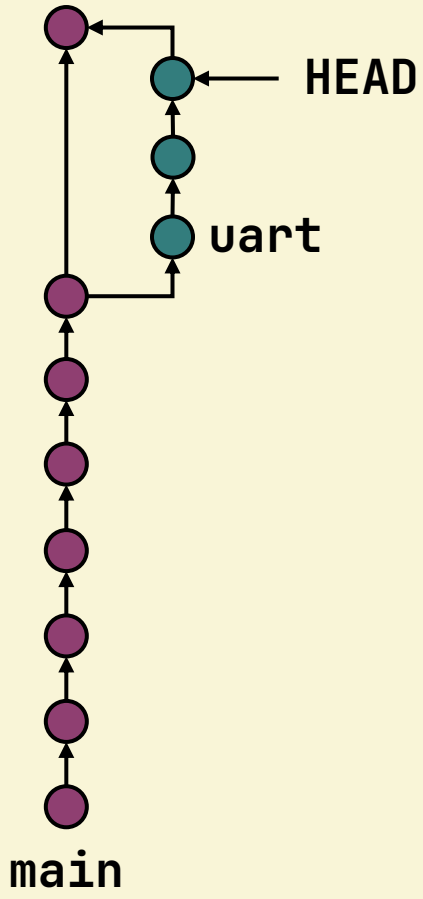

```
git checkout e3e1ead
```

Git Log



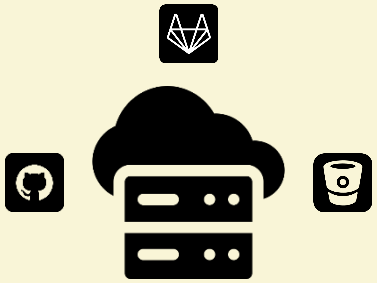

```
git checkout uart
```

Git Log




```
git checkout uart
```

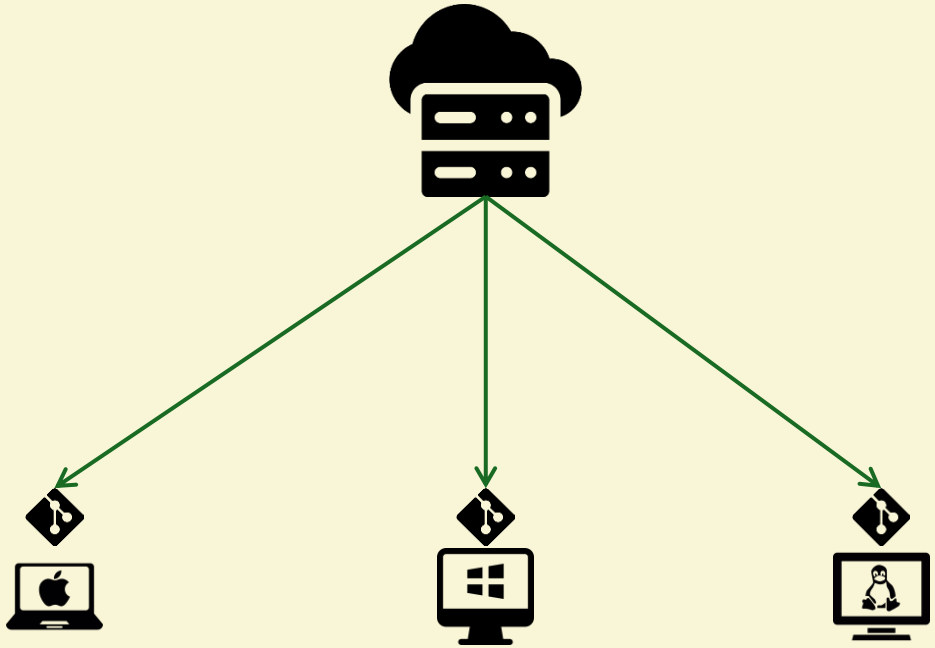
Repository



git clone <repo>

git clone <repo>

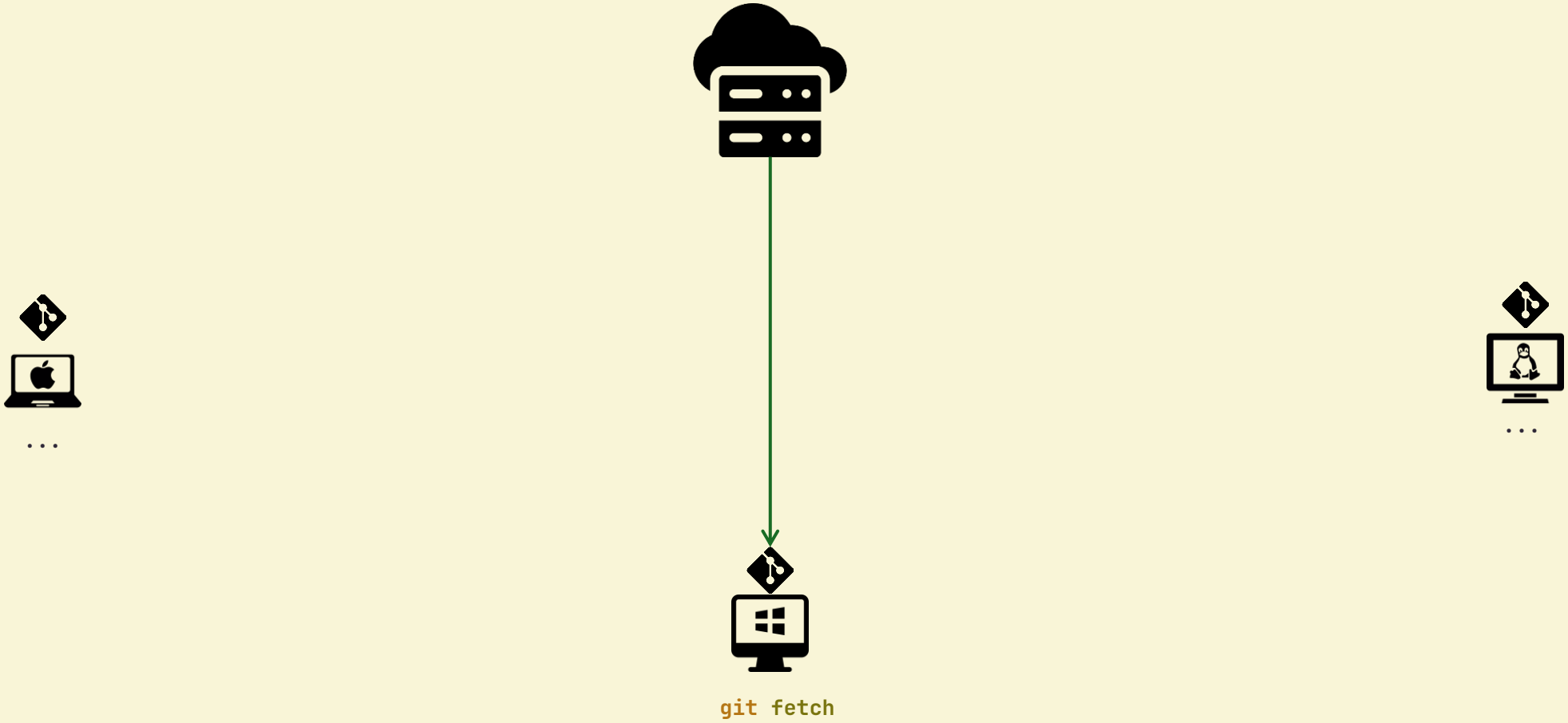
git clone <repo>

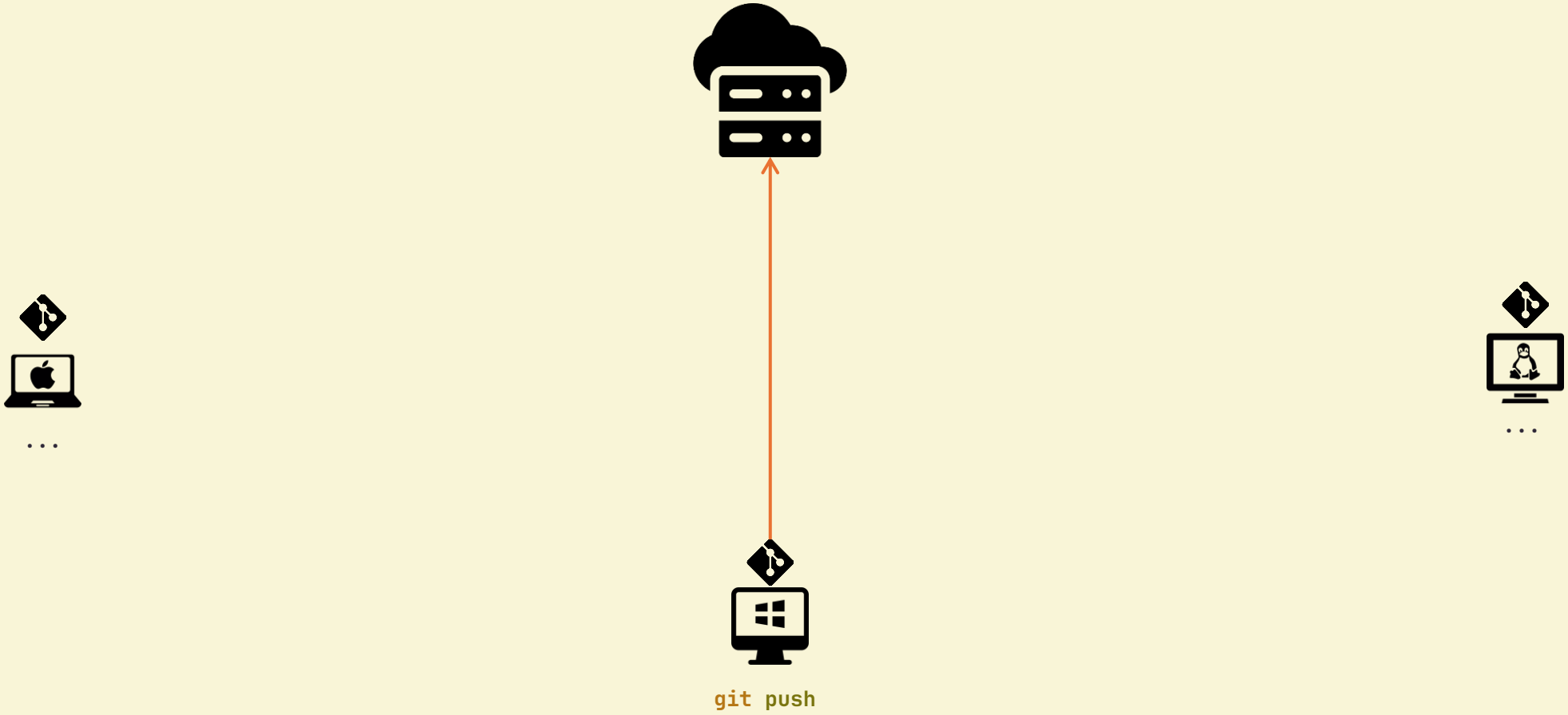


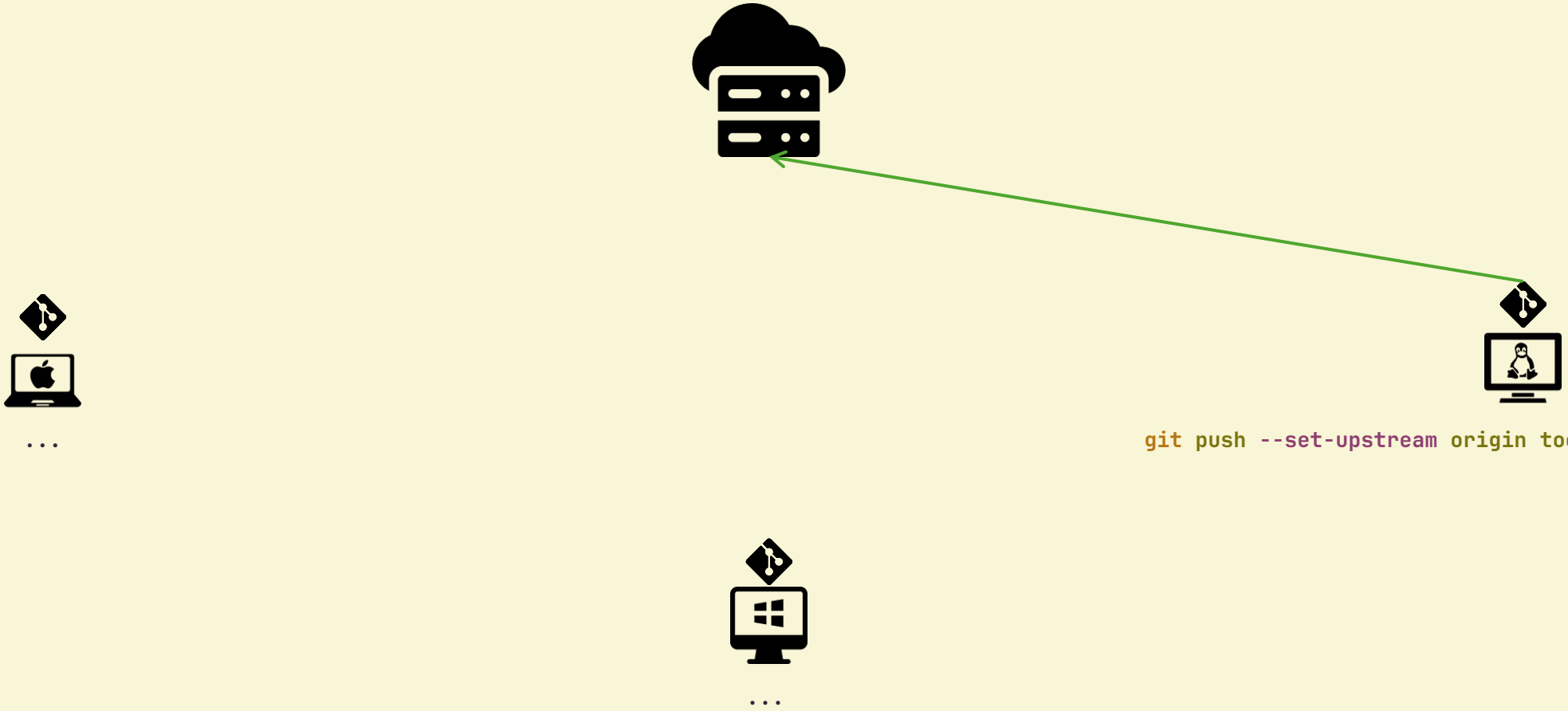
...

```
git checkout -b toolchain-experimental
```

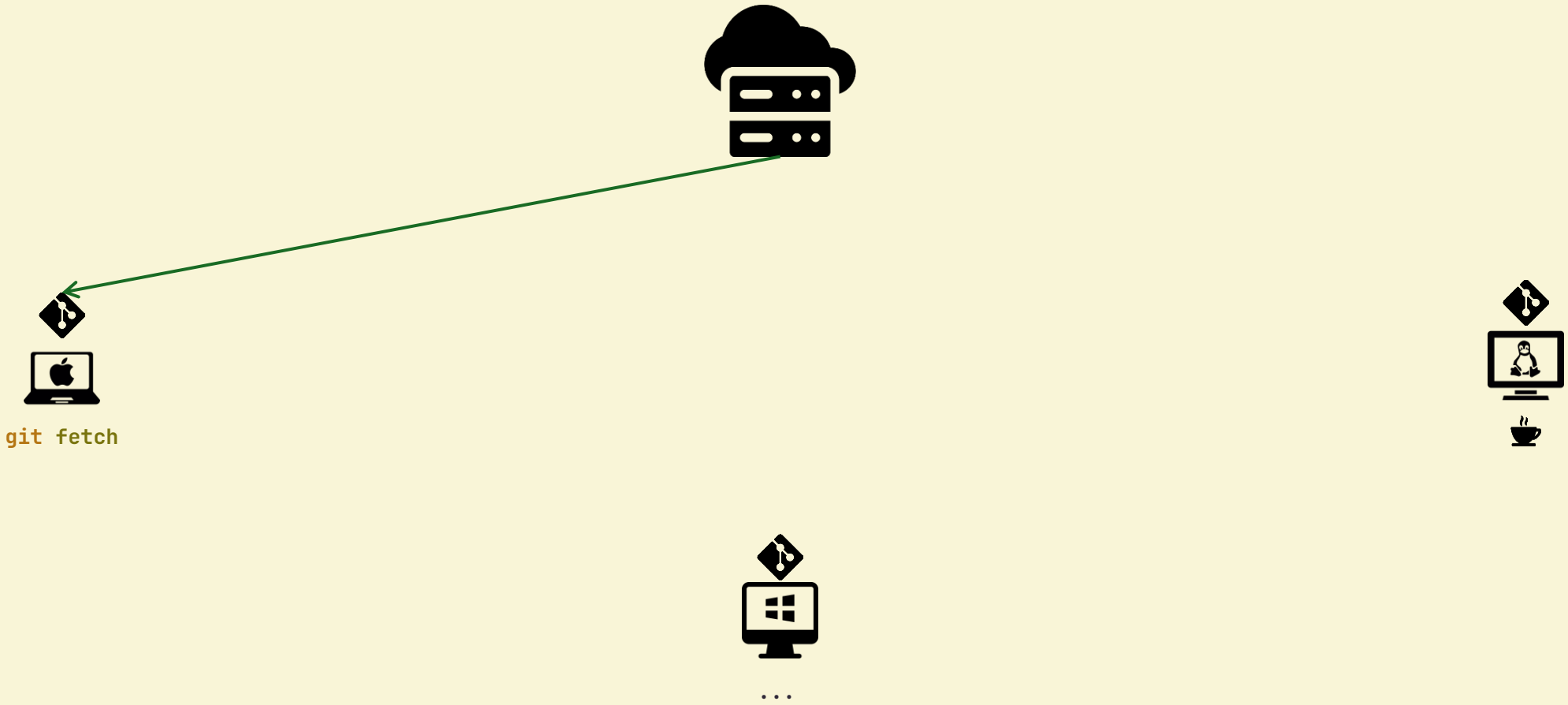
```
git commit -am "fix:..."
```


...
git fetch
...


...
git push
...

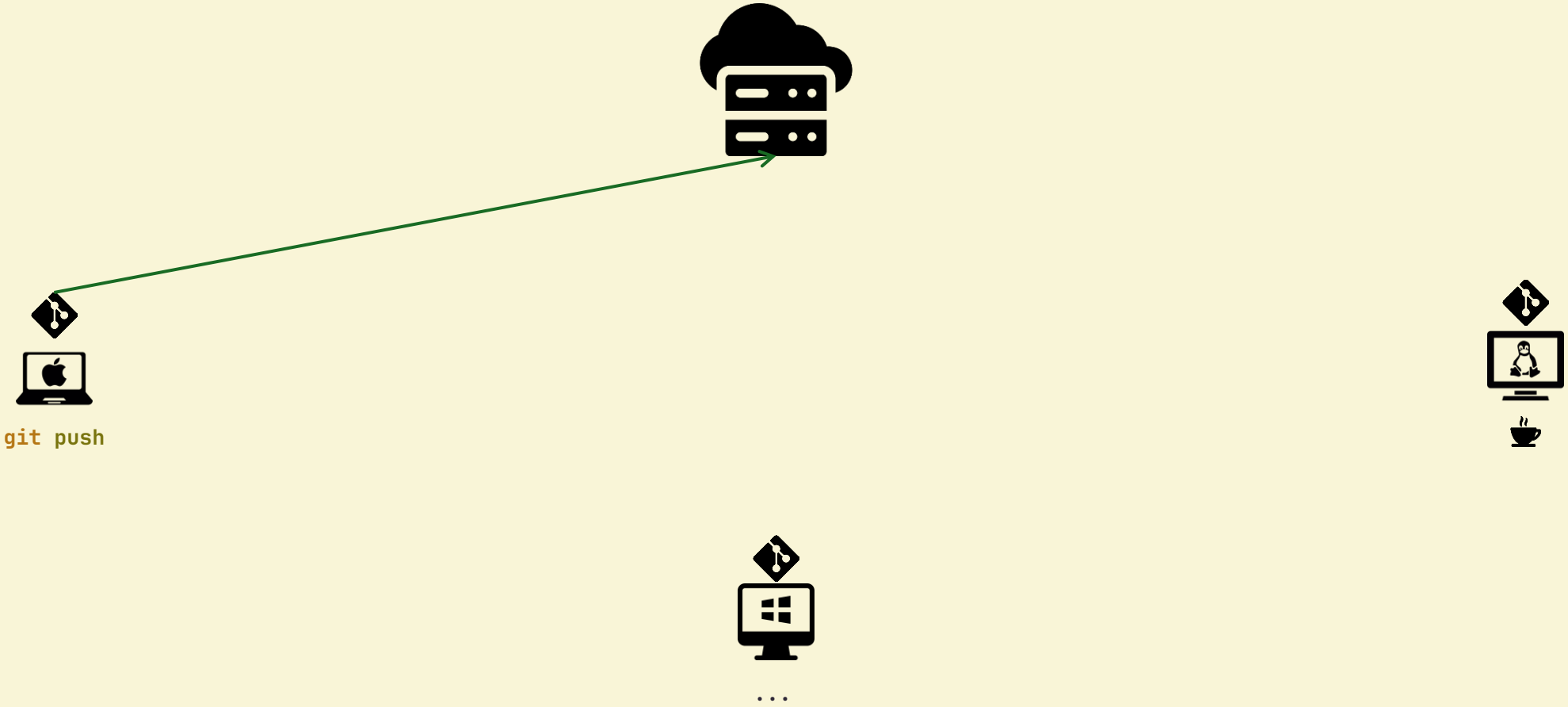


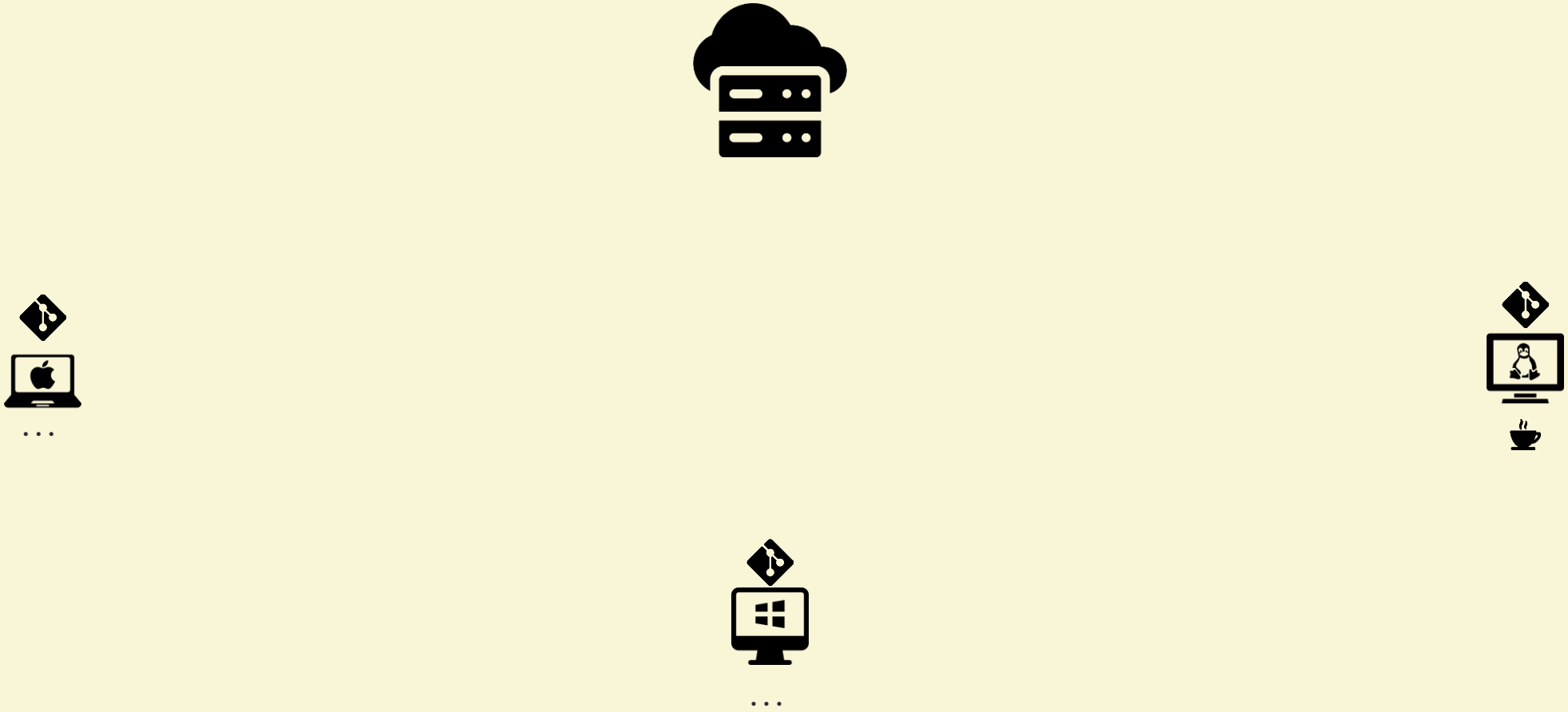

```
git push --set-upstream origin toolchain-experimental
```

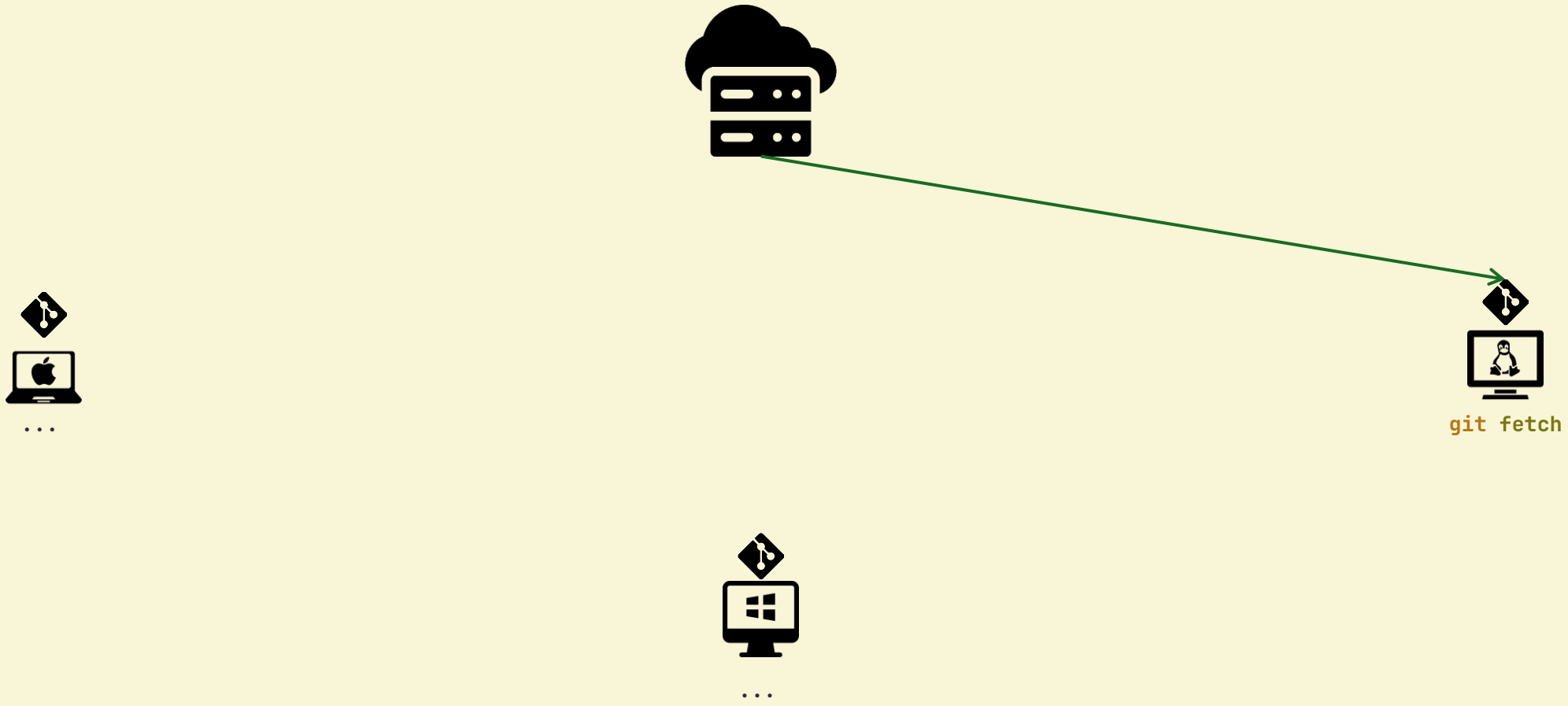

git fetch
...

```
git pull --rebase
```

...


git push
...

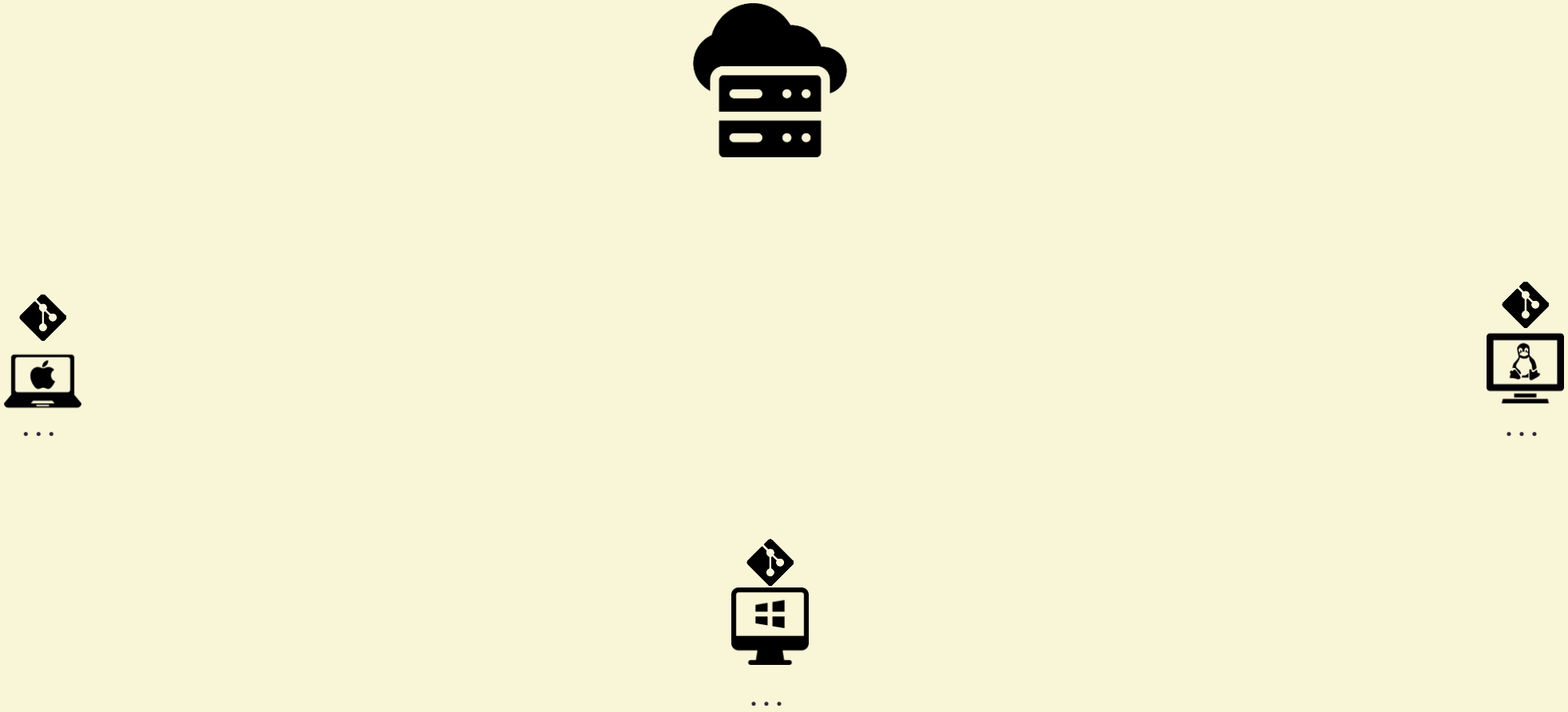



...
git fetch
...

...

git pull

...



Conflicting

```
git checkout dev
git commit -am "feat: improved robustness of gpio irq handler„
git checkout main
git commit -am "feat: use global gpio handler directly in the irq handler"
```

```
git merge dev

Auto-merging sw/hal/gpio.c
CONFLICT (content): Merge conflict in sw/hal/gpio.c
Automatic merge failed; fix conflicts and then commit the result.
```

```
git status -s

UU sw/hal/gpio.c
```

```
<<<<<<< HEAD
void gpio_irq_handler() {
    g_gpio->ints = g_gpio->regs->ints;

    if (gpio_irq_callback != 0) {
        gpio_irq_callback(g_gpio);
=======
void gpio_irq_handler(volatile gpio_handle_t *gpio) {
    gpio->ints = gpio->regs->ints;

    // IMPORTANT, DO NOT TOUCH!
    gpio->regs->ints = 0;

    if (gpio_irq_callback != NULL) {
        gpio_irq_callback(gpio);
>>>>>>> dev
    }
}
```

```
<<<<<<< HEAD
void gpio_irq_handler() {
    g_gpio->ints = g_gpio->regs->ints;

    // IMPORTANT, DO NOT TOUCH!
    gpio->regs->ints = 0;

    if (gpio_irq_callback != 0) {
        gpio_irq_callback(g_gpio);
=======
void gpio_irq_handler(volatile gpio_handle_t *gpio) {
    gpio->ints = gpio->regs->ints;

    if (gpio_irq_callback != NULL) {
        gpio_irq_callback(gpio);
>>>>>>> dev
    }
}
```

```
<<<<<<< HEAD
void gpio_irq_handler() {
    g_gpio->ints = g_gpio->regs->ints;

    // IMPORTANT, DO NOT TOUCH!
    g_gpio->regs->ints = 0;

    if (gpio_irq_callback != NULL) {
        gpio_irq_callback(g_gpio);
=======
void gpio_irq_handler(volatile gpio_handle_t *gpio) {
    gpio->ints = gpio->regs->ints;

    if (gpio_irq_callback !=) {
        gpio_irq_callback(gpio);
>>>>>>> dev
    }
}
```

```
<<<<<<< HEAD
void gpio_irq_handler() {
    g_gpio->ints = g_gpio->regs->ints;

    // IMPORTANT, DO NOT TOUCH!
    g_gpio->regs->ints = 0;

    if (gpio_irq_callback != NULL) {
        gpio_irq_callback(g_gpio);
    }
}
```

```c
void gpio_irq_handler() {
    g_gpio->ints = g_gpio->regs->ints;

    // IMPORTANT, DO NOT TOUCH!
    g_gpio->regs->ints = 0;

    if (gpio_irq_callback != NULL) {
        gpio_irq_callback(g_gpio);
    }
}
```

```c
void gpio_irq_handler() {
    g_gpio->ints = g_gpio->regs->ints;

    // IMPORTANT, DO NOT TOUCH!
    g_gpio->regs->ints = 0;

    if (gpio_irq_callback != NULL) {
        gpio_irq_callback(g_gpio);
    }
}
```

```
git status -s

UU sw/hal/gpio.c
```

```c
void gpio_irq_handler() {
    g_gpio->ints = g_gpio->regs->ints;

    // IMPORTANT, DO NOT TOUCH!
    g_gpio->regs->ints = 0;

    if (gpio_irq_callback != NULL) {
        gpio_irq_callback(g_gpio);
    }
}
```

```
git add . && git status -s

M  sw/hal/gpio.c
```

```
git commit \
-m  "merge: branch 'dev' into 'main'" \
-m "Combine robustness logic from 'dev' with global handler in the irq function"

[main 8f932e1] merge: branch 'dev' into 'main'
```

Ignoring

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
└── sw
    ├── build
    │   └── program.elf
    └── program.c
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
└── sw
    ├── build
    │   └── program.elf
    └── program.c
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
└── sw
    ├── build
    │   └── program.elf
    └── program.c
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
└── sw
    ├── build
    │   └── program.elf
    └── program.c
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore <EMPTY>

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/

# ignore cadence license key
sim/cadence.key
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/

# ignore cadence license key
sim/cadence.key
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/

# ignore cadence license key
sim/cadence.key

# ignore all pdk's
rtl/pdk/*
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/

# ignore cadence license key
sim/cadence.key

# ignore all pdk's
rtl/pdk/*
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/

# ignore cadence license key
sim/cadence.key

# ignore all pdk's
rtl/pdk/*
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       ├── .gitkeep
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/

# ignore cadence license key
sim/cadence.key

# ignore all pdk's
rtl/pdk/*
# keep the pdk folder itself
!rtl/pdk/.gitkeep
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       ├── .gitkeep
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

File: .gitignore

```
# simulation cashe
sim/sim_build/

# software build output
sw/build/

# ignore cadence license key
sim/cadence.key

# ignore all pdk's
rtl/pdk/*
# keep the pdk folder itself
!rtl/pdk/.gitkeep
```

```
.
├── rtl
│   ├── core.sv
│   ├── mcu.sv
│   └── pdk
│       ├── .gitkeep
│       └── tsmc_mem.sv
├── sim
│   ├── cadence.key
│   ├── restore.tcl
│   └── sim_build
│       └── cache.c
├── sw
│   ├── build
│   │   └── program.elf
│   └── program.c
└── .gitignore
```

Advancing

Stashing

```
# will save this for later
git stash push -m "wip-i2c-rw“

# going to check other branch quickly
git checkout main

# okay, let's go back
git checkout i2c-driver

# let's see witch one that was
git stash list

# restore stashed progress
git stash pop stash@{0}

# or to save in stash
git stash apply stash@{0}
```

Blaming

```
# who did this stupid breaking change???
git blame rtl/ibex_simple_system.sv
```

```
8401e312 rtl/ibex_simple_system.sv      (Rock Moose     2025-09-15 20:43:45 +0300 360)     `ifdef UART_NO_FIFO
8401e312 rtl/ibex_simple_system.sv      (Rock Moose     2025-09-15 20:43:45 +0300 361)     `else
2cf6f5ca rtl/ibex_simple_system.sv      (la_h1          2025-08-19 15:55:20 +0300 362)     .i_cts_n    (1'b0),
f2213f8c rtl/ibex_simple_system.sv      (4rcana         2025-08-08 16:45:03 +0300 363)     .o_rts_n             (uart_nrts_o),
f2213f8c rtl/ibex_simple_system.sv      (4rcana         2025-08-08 16:45:03 +0300 364)     .o_uart_rxfifo_int   (uart_rxfifo_int_o),
f2213f8c rtl/ibex_simple_system.sv      (4rcana         2025-08-08 16:45:03 +0300 365)     .o_uart_txfifo_int   (uart_txfifo_int_o),
8401e312 rtl/ibex_simple_system.sv      (Rock Moose     2025-09-15 20:43:45 +0300 366)     `endif
8401e312 rtl/ibex_simple_system.sv      (Rock Moose     2025-09-15 20:43:45 +0300 367)
8401e312 rtl/ibex_simple_system.sv      (Rock Moose     2025-09-15 20:43:45 +0300 368)     .o_uart_rx_int       (uart_rx_int_o),
8401e312 rtl/ibex_simple_system.sv      (Rock Moose     2025-09-15 20:43:45 +0300 369)     .o_uart_tx_int       (uart_tx_int_o),
f2213f8c rtl/ibex_simple_system.sv      (4rcana         2025-08-08 16:45:03 +0300 370)
1796ed94 rtl/ibex_simple_system.sv      (la_h1          2025-08-13 13:41:32 +0300 371)     // .i_uart_rx               (uart_rx_i),
c0203efa rtl/ibex_simple_system.sv      (la_h1          2025-08-13 13:24:58 +0300 372)     // .o_uart_tx               (uart_tx_o)
fb6f346c rtl/ibex_simple_system.sv      (dovydasliutkus 2025-10-16 13:47:53 +0300 373)     .i_uart_rx    (ext_pad_i[0]),
a0d9c560 rtl/ibex_simple_system.sv      (dovydasliutkus 2025-11-04 18:18:27 +0200 374)     .o_uart_tx    (uart_tx_o)
```

Other

```
# I remember that working last week...
git bisect

# no, no, no, license key is added to the commit!
git filter-repo

# let's add just this write function
git add -p

# where is that deleted branch?
git reflog

# multiple repositories in larger projects
git submodule
```

ktu

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T171B101 Lustų ir jų sistemų prototipavimo, gamybos ir testavimo technologijos